

Self-Information (Surprisal)

math-foundations / 35-self-information

First, an example

Before formalising surprisal, fix intuition with coins. A *fair* coin has $p(H) = p(T) = 0.5$, so its self-information in bits is $I(H) = -\log_2(0.5) = 1$ bit — one bit of surprise per flip, no more, no less. *Read aloud:* “I of H equals minus log base two of one-half equals one bit.”

Now bias the coin: $p(H) = 0.9$, $p(T) = 0.1$. Then $I(H) = -\log_2(0.9) \approx 0.152$ bits (a heads is barely surprising, since we expected it), while $I(T) = -\log_2(0.1) \approx 3.32$ bits (a tails is genuinely informative). Rare outcomes carry more surprisal: this is the whole picture.

Finally, flip the fair coin twice and observe HH . Independence gives $I(HH) = I(H) + I(H) = 2$ bits — surprises *add* for independent events. The rest of this lesson explains why $-\log p$ is the unique function with these properties.

1 Motivation

Information theory begins with a deceptively simple question: how much *information* does the occurrence of a single event convey? Intuition gives three desiderata.

- A certain event ($p = 1$) conveys no information.
- Rarer events convey more information than common ones.
- Two independent events convey, jointly, the sum of their individual informations.

Shannon (1948) showed that these three properties, together with continuity, single out a one-parameter family of functions: $I(p) = -c \log p$. The constant $c > 0$ merely chooses units; up to that choice, $-\log p$ is the *unique* measure of surprisal.

2 Definitions

Definition 1 (Self-information / surprisal). Let X be a discrete random variable with probability mass function p on a countable alphabet $\mathcal{X}$, and let $x \in \mathcal{X}$ satisfy $p(x) > 0$. The *self-information* of the outcome x is

$$I(x) = -\log p(x).$$

Read aloud: “I of x equals minus log of p of x.” The base of the logarithm fixes the units of I :

$$\log_2 \Rightarrow \text{bits}, \quad \ln \Rightarrow \text{nats}, \quad \log_{10} \Rightarrow \text{hartleys}.$$

Read aloud: “log base two gives bits, natural log gives nats, log base ten gives hartleys.”

Definition 2 (Self-information of an event). For an event A with $\Pr(A) > 0$, define $I(A) = -\log \Pr(A)$. When $A = \{X = x\}$ this recovers the previous definition.

Remark 3. Self-information is a function of p , not of x in any geometric sense; two outcomes with the same probability carry the same surprisal regardless of their identity.

3 Additivity for independent events

Theorem 4 (Additivity). *Let A and B be independent events with probabilities $p_A, p_B \in (0, 1]$. Then*

$$I(A \cap B) = I(A) + I(B).$$

Read aloud: “ I of A intersect B equals I of A plus I of B .”

Proof. By the definition of independence,

$$\Pr(A \cap B) = p_A p_B.$$

Read aloud: “probability of A intersect B equals p sub A times p sub B .” Applying the definition of self-information,

$$I(A \cap B) = -\log \Pr(A \cap B) = -\log(p_A p_B).$$

Read aloud: “ I of A intersect B equals minus log probability of A intersect B equals minus log of p sub A times p sub B .” Using the homomorphism property of the logarithm, $\log(xy) = \log x + \log y$ for $x, y > 0$, we obtain

$$-\log(p_A p_B) = -\log p_A - \log p_B = I(A) + I(B). \quad \square$$

Read aloud: “minus log of p sub A times p sub B equals minus log p sub A minus log p sub B equals I of A plus I of B .”

Remark 5. Independence is essential. If $B = A$, then $A \cap B = A$ and $I(A \cap B) = I(A) \neq 2I(A)$ in general. The correct generalisation to dependent events introduces *conditional* self-information $I(B | A) = -\log \Pr(B | A)$, yielding the chain rule $I(A \cap B) = I(A) + I(B | A)$.

4 Uniqueness of $-\log p$

The additivity property is, in fact, almost sufficient to characterise surprisal.

Theorem 6 (Shannon uniqueness, abridged). *Let $f : (0, 1] \rightarrow \mathbb{R}_{\geq 0}$ be continuous, strictly decreasing, and satisfy*

$$f(pq) = f(p) + f(q) \quad \text{for all } p, q \in (0, 1].$$

Read aloud: “ f of p times q equals f of p plus f of q for all p, q in the half-open interval zero to one.” Then there exists a constant $c > 0$ such that $f(p) = -c \log p$.

Sketch. Set $g(t) = f(e^{-t})$ for $t \in [0, \infty)$. The functional equation becomes

$$g(s + t) = f(e^{-(s+t)}) = f(e^{-s}e^{-t}) = f(e^{-s}) + f(e^{-t}) = g(s) + g(t),$$

Read aloud: “ g of s plus t equals f of e to the minus s plus t , equals f of e to the minus s times e to the minus t , equals f of e to the minus s plus f of e to the minus t , equals g of s plus g of t .” i.e. g is additive on $[0, \infty)$. Together with continuity, Cauchy’s functional equation gives $g(t) = ct$ for some constant c . Strict monotonicity of f forces $c > 0$. Translating back, $f(p) = g(-\log p) = -c \log p$. $\square$

5 Worked example

Example 7 (Biased coin). Let $X \in \{H, T\}$ with $\Pr(X = H) = 0.9$, $\Pr(X = T) = 0.1$. Working in bits:

$$I(H) = -\log_2(0.9) \approx 0.152, \quad I(T) = -\log_2(0.1) \approx 3.322.$$

Read aloud: “I of H equals minus log base two of nine-tenths, approximately zero point one five two; I of T equals minus log base two of one-tenth, approximately three point three two two.” The expected self-information is

$$\mathbb{E}[I(X)] = 0.9 \cdot 0.152 + 0.1 \cdot 3.322 \approx 0.469 \text{ bits},$$

Read aloud: “expectation of I of X equals zero point nine times zero point one five two plus zero point one times three point three two two, approximately zero point four six nine bits.” the Shannon entropy of X , which is strictly less than the maximum of 1 bit achieved by a fair coin.

6 Connection to machine learning

For a language model $q(x_t \mid x_{<t})$, the per-token cross-entropy loss

$$\mathcal{L}_t = -\log q(x_t \mid x_{<t})$$

Read aloud: “script L sub t equals minus log q of x sub t given x less than t.” is precisely the self-information that the model assigns to the observed token. Minimising average loss is therefore minimising mean surprisal under the model. Perplexity, $\exp(\overline{\mathcal{L}})$, is the exponentiated mean surprisal — the geometric-mean inverse-probability per token.